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Adapting a Transdisciplinary Approach to Regional Development in the Case of Facilitating Planning of Energy Systems

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Abstract. In this paper, we aim to facilitate regional development through collaborative meta-governance processes, involving municipalities, energy companies and more actors. The case in point involves increased electrification through the development of local energy systems in a sub-region comprising 15 municipalities and 21 grid owners and is an exemplary case where a transdisciplinary engineering approach is not only suitable but the only path forward. The complex problem landscape comprises interdependencies across different roles, such as politicians, civil servants, and engineers at energy companies, where autonomous entities act independently. We employ a design science research approach to create artefacts to support the meta-governance mechanisms needed to accelerate social change processes. One artefact is a system dynamics simulation model to analyze scenarios considering the electrification of vehicles and implementing large wind and solar energy units to enable the establishment of new industries. We provide brief overviews of how three artefacts assist in visualizing 1) roles, 2) structures, and 3) scenarios to the decision-makers, to facilitate various transdisciplinary decision-making processes in regional development. In the discussion, we synthesize our learnings into a model to support mitigating powerlessness in this complex multi-stakeholder context. Finally, we lay out future research to further contribute to the social change and regional development we believe is necessary.

Keywords. Regional development, Meta-governance, Planning of energy systems, System dynamics simulation, Transdisciplinary engineering.

Introduction

The transition away from a fossil fuel dependent society is a global endeavour, based on the UN system: The Intergovernmental Panel on Climate Change and the Agenda 2030. The European Union is working toward the goal of being climate neutral in 2050, the current objective being a reduction of the emissions of greenhouse gases by 55% before the year 2030 [1]. Sweden is almost an outlier when it comes to the use of renewable energy sources within the EU, having a 62,6% share of renewables in 2021 [2].

The Swedish Energy Agency regularly publishes forecasts on the need for electricity in Sweden, where the yearly consumption is predicted to double from approximately 150TWh to more than 300TWh [3]. In West Sweden, different scenarios point toward a

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double demand, from 20TWh to 40TWh or more, per year, in the upcoming years. Two transitions drive this increased demand: the transportation sector and the industry [4].

Picchi et al [5] point out that the sustainability of the energy transition implementation process is affected by a lack of social-ecological systems thinking and that future development of sustainable energy landscapes requires informed decision-making.

Regions are central actors in this societal change and are expected to drive different kinds of development and compete for human capital and other resources [5]. Regional development is in the Swedish context at the core of a region's responsibilities, mandated by law. Strategies for regional development focus on sustainable development, transition, and recent electrification.

Traditional policymaking, reliant on public actors gathering expertise and implementing regulatory solutions, struggles to address modern societal complexities [6]. Policymakers express powerlessness with regards to this. These challenges demand a governance approach that acknowledges uncertainties, interconnectedness, and potential consequences. Historically, regional policy focused on economic growth and reducing disparities, but evolving strategies require a more integrated approach. Effective regional policy necessitates collaboration across organizational as well as economic, social, and cultural realms, involving actors in decision-making processes [7]. Development is now viewed as transformative, emphasizing holistic behaviour and feedback mechanisms to avoid ineffective policymaking [7].

System dynamics, applied as a simulation approach in our research, is seen as a robust and powerful tool for describing and analyzing the intricate interconnections within complex systems, enabling the simulation of system behaviour across time [8]. It is used to understand and evaluate the impacts of variables within renewable energy systems and as a decision-support tool across diverse sectors. In their paper [9], Saavedra et al present the latest trends in system dynamics related to the energy sector and outline four primary themes: Fossil Fuels, Renewable Energy, Electricity, and other energy-related resources. The research community has also frequently utilized system dynamics within areas such as regional planning [10], [11], as well as energy policy development and energy system modelling and decision support [12], [13], [14], [15].

Contributions

To address the above, this research aims to enable effortless and more productive planning, decision making and implementation where multiple actors integrate their efforts to achieve an impact on the speed and quality of the transition through a transdisciplinary approach combining technology and social sciences. Contrasting previous researchers mentioned above, this paper aims to combine both strategic planning and energy policy-making decision support by utilizing system dynamics simulation. The stated contribution of the research is enabling more efficient facilitation of regional development and transition.

The paper contributes initial interpretations of the transition of energy systems for the studied region in West Sweden and one of its sub-regions through the lenses of the theory identified.

1. Theoretical frameworks – Meta-governance and powerlessness

In the theoretical frameworks for the conducted research, there is a need to integrate perspectives by studying meta-governance and conditions for powerlessness. Gjaltema et al. [16, p. 177], define meta-governance as ”a practice by (mainly) public authorities that entails the coordination of one or more governance modes by using different instruments, methods, and strategies to overcome governance failures”. Systems of powerlessness appear in Rosa's thinking, where policymaking is described as futile. "It seems that … it has become politically impossible to plan and shape society over time; the time of political projects, it seems, is also over" [17, pp. 21–22].

More concretely, Gjaltema et al. [16] explore the "who, what, why and how" of meta-governance where combining the who and what lead to four ideal types of meta-governance: network meta-governance, multilevel meta-governance, meta-governance of multiplicity and meta-governance of modes. Transitioning to meta-governance entails new roles for government, as well as networks combining "control and facilitation" [18]. Regional development is well suited for meta-governance, even though challenges exist [19]. Hooge et al [20, p. 1595], present strategies for network governance in a heuristic analytical framework, distinguishing: 1) Network design strategies, 2) Resourcing strategies, and 3) Framing strategies.

A transition from a modern to a post-modern context has taken place in recent years, impacting decision-making processes. Foucault [21], essentially argued that where modern democratic and bureaucratic institutions see themselves as rational, almost scientific, this is not the case.

The German sociologist Hartmut Rosa [17], grapples with modernization and presents "acceleration" as a prerequisite temporal dimension for understanding the process of modernization, which takes place in three forms 1) Technological acceleration in goal-directed processes speeding up communications, production and more; 2) Acceleration of social change where rates of change in society themselves are changing; 3) Acceleration of the pace of life. Three drivers of acceleration are identified: 1) The economical motor, where acceleration is driven by capitalism; 2) The cultural motor, where acceleration is presented as a strategy to mitigate expectations of a fulfilled life, filled with realizing as many options as possible, and the fact that time is limited; 3) The structural motor, where the principle of functional differentiation drives acceleration.

The systems of powerlessness, mentioned above, impact decision-making and politics in an overwhelming way. Rosa points out: "As a result, politics, too, has become 'situationalist': it confines itself to reacting to pressures instead of developing progressive visions of its own. Very often, political decisions no longer aspire to actively steer (acceleratory) social developments, but are defensive and decelerator" [17, p. 21]. According to Rosa [17, p. 14], complexity is ever-increasing, due to a change in societal structure moving away from hierarchies, from government to governance. Moreover [38, p. 14], Rosa concludes that the "structural problem at the heart of this disappearance of politics is the political system's fundamental inability to accelerate" [17, p. 22].

In our research, enhanced capacity for meta-governance through, among others, the artefacts presented below is a way to mitigate powerlessness.

2. Method

In our research, we apply a pragmatic approach which "allows the possibility of choosing the appropriate research methods from the wide range of qualitative and/or quantitative methods" [22, p. 12], The addressed problem, of facilitating development in a complex and multi-stakeholder context, clearly benefits from a transdisciplinary engineering approach to address both the technical aspects but foremost the social collaborative aspects of the problem. Furthermore, a design-science research approach aims to propose a "purposeful organization of resources to accomplish a goal" [23, p. 78], in our case developing artefacts supporting meta-governance.

The artefacts presented in this article have been introduced, and subsequently adapted, in a multitude of settings including many kinds of actors and stakeholders such as policymakers, energy companies and civil servants. This process aligns with Hevner [23, pp. 88, 89], describing the design as an iterative search process.

Moreover, visualization is applied as a pedagogical tool to approach complex problems, where simulation can enable learning to understand the consequences of decisions [24]. At the same time, any visualization and modelling build on reduction [25], and a perceived understanding might be false. In our case, the produced artefacts serve as visualizations to facilitate a deeper understanding of different stakeholders' perspectives by clarifying the roles and responsibilities between energy companies and municipalities and supporting the governance structures, where applying system dynamics simulation has enabled scenario planning of possible challenges, facilitating dialogue and a collaborative approach.

The board of the sub-regional Association of Municipalities has approximately 10 regular meetings per year. In these meetings, our research has collected empirical foundations and delivered artefacts. One of the authors is a strategist with the organization, and regularly introduces new information and perspectives to the policymakers, advancing strategic discussions and decision-making.

3. Empirical background

Swedish electrical grids are divided into three tiers: 1) the backbone grid (TSO), owned by the state; 2) The regional grids, mainly owned by private companies, but also by companies at least partly owned by the state; and 3) The local distribution grids (DSO) where consumers apart from really large industries are connected, oftentimes owned by municipalities or cooperatives of residents, or in some rare cases privately owned.

In Sweden, there is a political debate on which power sources are needed and should play a role in the transition. After some debate within the board, nuclear energy was also included moving forward [26]. This reflects the change on the national level, where the current government is focusing heavily on nuclear energy to facilitate electrification.

Both the government, the regions and municipalities have ambitions concerning economic growth, new jobs, and increasing the capacity of the grids and energy production. This creates a complex context for decision-making with multi-level governance as well as contradicting policies. Different actors have different incentives for contributing to the desired development. Municipalities, for instance, anticipate a positive demographic development, strengthening their tax base and providing financing for welfare. Grid owners might fear huge investments to enable electrification. Hence, the situation is characterized by a complex multi-stakeholder context.

All decision-making bodies in the public sector rely on cooperation between civil servants such as spatial planners and politicians. Roles and mandates between them are regulated in law. However, the exact delimitations are locally or regionally negotiated and vary a lot between authorities. A common metaphor for the distribution of mandate is how versus what, where politicians are supposed to decide what needs to be done and civil servants then identify and carry out the how.

With increasing levels of complexity entailing all parts of life, leaving a more linear reality behind, a new context for decision-making where both intended and unintended consequences are harder if at all possible to predict emerges [24] [25]. Policymakers have often expressed a sense of powerlessness facing this complexity. This puts strain on the relationships between civil servants and politicians. Civil servants can no longer produce an unambiguous basis for decisions, but rather present different alternatives with tentative descriptions of consequences as a basis for decisions, prioritizing and outlining conflicts of interest. Often policymakers ask for more "concrete" proposals to mitigate this. They request reduction (simplicity) often phrased as a demand for more concrete presentations and alternative solutions. This has consistently been the case within the studied region while working to create coherent and relevant strategies for the transition of energy systems.

During the recent years, 2021 through 2023, the transition of the energy systems and more specifically electrification has been a recurring part of the conversations and decisions [26]. Discussions have ranged from charging for electrical vehicles (EVs) to more recently establishing a sub-regional plan. Moreover, the focus of the discussions has changed from initially considering requests for more capacity in the transmission network to considering local energy production, supporting higher self-sufficiency. This is driven by neo-industrialization and the prospect of more taxpayers.

4. Results

Roles and governance structure

Two artefacts to mitigate the challenges have been developed to achieve more effortless planning of energy systems, one concerning role, depicted in Figure 1, and one concerning governance structures, depicted in Figure 2.

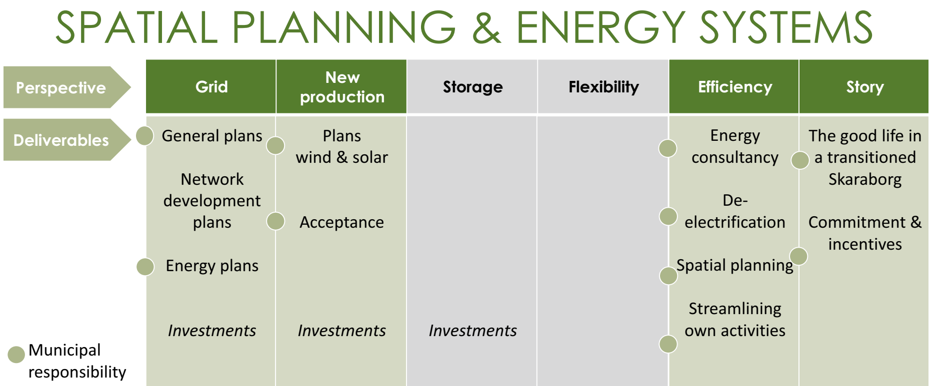

Figure 1. Roles and responsibilities between energy companies and municipalities.

It has recently become obvious, when the case of electrification governance has evolved, that clarity with regards to roles and responsibilities is crucial. At the beginning of the transition work, this was not clear, creating unnecessary friction between different actors and stakeholders, in fact even supporting the hypothesis of powerlessness. The artefact depicted in Figure 1, has led to efforts and leadership focusing on the parts of electrification that different actors can actually influence, creating a smoother environment and also moving work forward. The green columns in the illustration are the ones to which municipalities can contribute, and the grey represents actions that other actors are responsible for, e.g. investments. For instance, 13 out of 15 municipalities in the sub-region are currently establishing energy plans, and all relevant municipalities are establishing plans for wind energy enabling new local production of electricity.

Employing strategies proposed by Hooge et al. [20], the roles of municipalities and energy companies are separated. The regional association of municipalities has also been able to provide financing for facilitating development between energy companies. Different types of energy companies have different incentives for contributing to the development, whereas municipality-owned companies can be expected to be governed to contribute to regional development.

The "division of labour", between the municipalities and the energy companies, enabled the establishment of a structure for governance with a framework for co-creation and coordination, depicted in Figure 2. This structure has yet to be fully materialized, due to organizational constraints within the regional association.

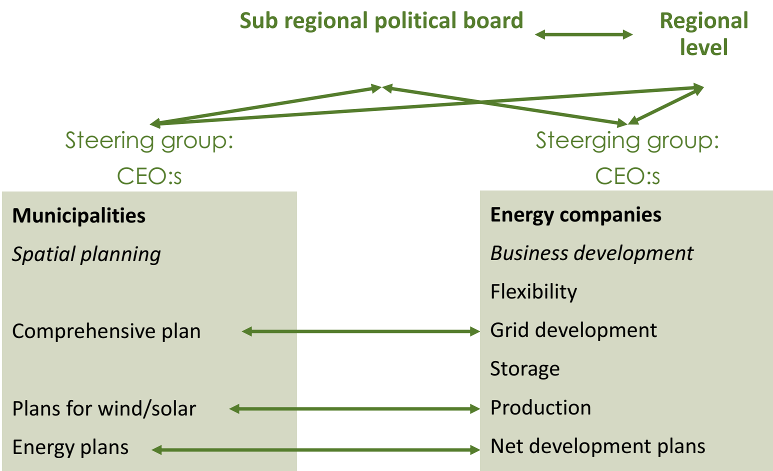

Figure 2. Structures for governance and interaction/integration.

Most actors in the regional system agree that lack of electricity prevents desired development. Several initiatives are underway where for example the Region is working with TSO and regional grid owners are identifying and potentially eliminating bottlenecks in the national and regional grids. The regional association of municipalities promotes energy planning by its members, as well as planning for wind power. A decision to establish a sub-regional plan for energy supply is also in place. All of these, and other initiatives are interdependent where, as an example, adding new local production demands a developed grid, large-scale storage, and flexibility solutions.

Establishing structures includes establishing who is the meta-governor. In this case, both the region and the regional association of municipalities act as meta-governors. However, one observation is that the role of the meta-governor in the system above is not clear. This complicates the facilitation and coordination of the desired development. The arrows in Figure 2 do not represent formal power, but rather capture a need for co-creation. Moreover, a clear sense of urgency, with the threat of loss of jobs and more, creates a situational context as outlined by Rosa [17] characterized by reacting to pressures rather than actively accelerating the social change deemed necessary.

Simulation model and results

The system dynamics simulation model was created to assess the balance between energy production and consumption in two municipalities within the studied sub-region, accounting for the implementation of various policies. Generally, we followed Sterman's [24] approach in modelling and testing the resulting models. As foundational input data, a huge data set [27] was used, provided by the government through its county boards outlining municipalities' energy-production and -consumption patterns from 2020, which is the most recent up-to-date data. The data has been used to visualize each municipality's energy balance through Sankey diagrams. The aim was to build a simulation model to visualize how various scenario implications decision-makers could face, drive changes to the energy balances using Sankey diagram representations, and ultimately identify appropriate policy variables. The six identified transitional scenarios were the following:

1) Increased ratio of electrified vehicles in the vehicle fleet
2) Establishment of new larger industries
3) Addition of new households
4) De-electrification by transitioning households to district heating instead of electricity
5) Implementation of larger solar parks
6) Installation of large wind power units

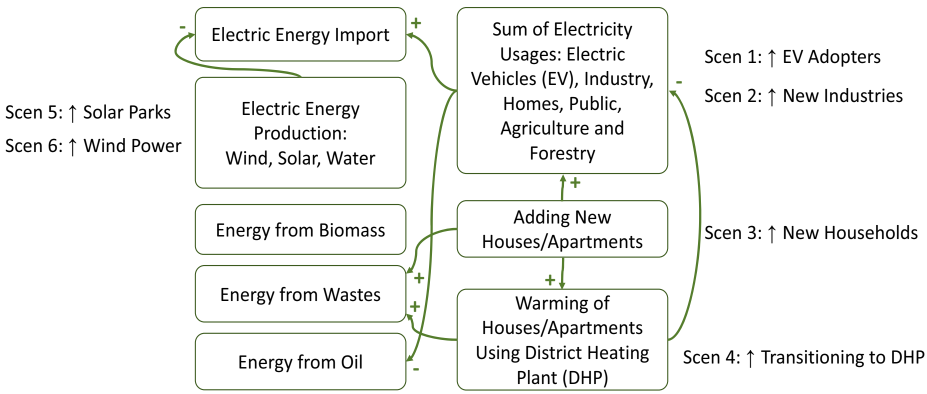

Figure 3. An abstraction of the system dynamics simulation model and tested scenarios.

Figure 3 depicts an abstraction of the simulation model to ease readability. To the left are the various energy sources, where own electric energy production proportionally reduces the electric energy import, and to the right are the main consumers illustrated. The transitional scenarios, "Scen 1-6", manipulate the dynamics of the model according to the following.

In scenario 1, the adoption of EVs until the year 2030 is captured, reducing the fossil fuel dependency on oil for transports. At the same time, this increases the need for electric energy production supplied by increasing the electric energy import.

In scenario 2, introducing new industry establishments (NIEs), based on a power need of 16MW, each NIE generates an increased yearly energy load of 86GWh. The introduced load increases the sum of electricity usages from Industry, leading to higher electric energy import.

Scenarios 3 and 4 introduce expected changes in total energy consumption from new households, such as new houses and apartments. Depending on policy parameters for the ratio of how these households should have to use district heating or not can be explored and evaluated towards the energy consumption and corresponding production needs through increased electric energy import and increased use of wastes for the cogeneration plant if utilized in the local grid, or other type of district heating technology.

Scenarios 5 and 6 introduce locally produced electric energy, which can be increased by building large solar parks, producing 1GWh annually, and large wind power units, producing 17.3GWh annually. This local electric energy production correspondingly reduce the need for electric energy import after an implementation delay.

To highlight some findings, we explore the scenarios for one of the municipalities covering a local grid in the region, and we also explore how collaboration across municipalities, involving in total of 5 local grids, could support the electrical transition. We especially focus on line 8 in both graphs. Figure 4 depicts the combined scenario for the *local* grid in the municipality, implementing all challenges and local productions. Analysing the results indicates insufficiency to self-supply energy to support any NIE when considering one local grid. In contrast to Figure 5, which depicts a *collaboration* effort representing 5 local grids joined forces, in the combined region scenario, how two NIEs could be manageable if, indicating we need to collaborate across existing boundaries to succeed. However, this type of collaboration is currently restricted by jurisdiction, in need of being confronted and changed, where any distribution of electric energy across the local grids or back to the regional grid is prohibited. These findings are in support of facilitating an explorative, progressive dialogue to open up for the required changes. In Figure 6, an illustrative example is depicted of how the resulting Sankey diagram for a *local* scenario was impacted from a selected future scenario.

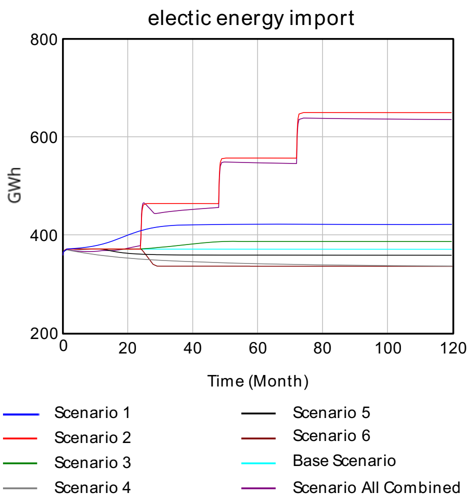

Figure 4. Need to import electricity *locally*.

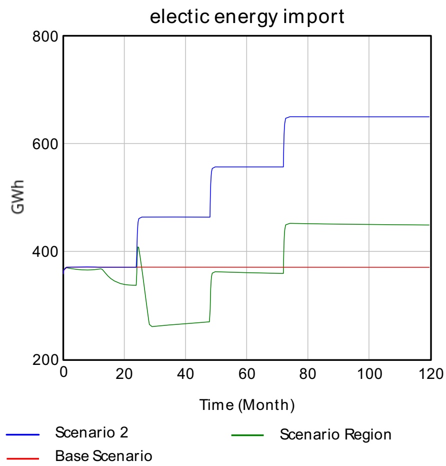

Figure 5. Need to import electricity *collaboration*.

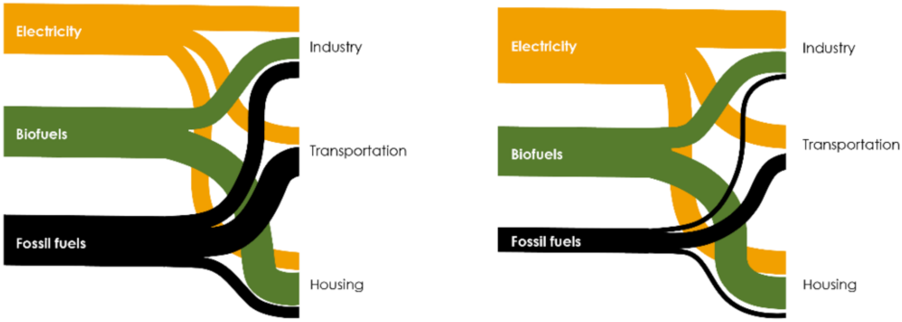

Figure 6. A simplified illustrative example of how Sankey diagrams of energy-production patterns (from electricity, biofuels, and fossil fuels) and -consumption patterns (by industry, transportation, and housing) was used. The left Sankey diagram represents a current state and the right one the possible future state, summarizing how the increased demands for electrical energy is replacing oil use and fossil fuel dependencies for a certain *local* grid.

5. Conclusions

The transition away from fossil fuel dependency towards sustainable energy is imperative. The discourse surrounding energy transition has garnered significant scholarly attention, emphasizing its multifaceted nature beyond technological advancements, necessitating comprehensive changes across economic, political, institutional, and socio-cultural spheres. The transdisciplinary approach presented in this paper showed the importance of collaborative meta-governance processes in fostering regional development to mitigate powerlessness. Where the engagement of diverse stakeholders, including municipalities, energy companies, and various actors through a structured meta-governance process address the intricate challenges posed by the electrification of local energy systems across a complex landscape of 15 municipalities and 21 grid owners. In addition, the development and utilization of a system dynamics model and simulation of "what if"-scenarios enabled visualization of the impacts of energy transition for the stakeholders, particularly concerning the integration of renewable energy sources and the emergence of new industries. This resulted in expanded dialogue to elaborate on long-term solutions.

Thus, though the transdisciplinary approach informed discussions among stakeholders was fostered, it guides the identification of actions necessary to expedite desired societal transformations.

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